

Sumit Kumar

4th Year Undergraduate

Department of Computer Science and Engineering

Indian Institute of Technology Kharagpur

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in Sumit Kumar SumitKumar-17

Academic Qualifications

Year	Degree/Certificate	Institute	CGPA/%
2022 - Present	5 year Dual Degree Course(BTech. + MTech.) in Computer Science	Indian Institute of Technology, Kharagpur	8.82/10
2022	Class XII (CBSE)	Delhi Public School, Kalyanpur	94.20%
2020	Class X (CBSE)	Delhi Public School, Kalyanpur	96.60%

Work Experiences

Software Engineer Intern - Atlassian

May'26-Jul'26

- Shipped **Bulk Archive** and **Bulk Delete Teams** across all Atlassian orgs via **GraphQL mutations** and frontend dialogs
- Replaced a **3-4s manual deletion flow** with one API call using hierarchical async deletion on **AWS SQS** and **DynamoDB**
- Proposed a fix for **notification spam** in Bulk Archive/Delete and validated it via **Splunk** in staging and production
- Instrumented API latency, failure, and counter metrics with **SignalFx** dashboards and **Terraform**-managed alert rules

Backend Engineer Intern - Steps AI

Jun'25-Present

- Engineered async **Python microservices** with **FastAPI** and **Pydantic**, serving agent, copilot, voice, and FAQ APIs
- Orchestrated **LangGraph** and **LangChain** agents with streaming, checkpointing, and **Milvus/Cohere** retrieval
- Built a **DAG-based workflow platform** on **PostgreSQL/Redis** with retries, conditional steps, and connector actions
- Automated **Shopify/WooCommerce** product sync and embedding ingestion via **Temporal** workflows across **S3** and **Milvus**
- Deployed services with **Docker** and **ArgoCD** GitOps, adding **Sentry/OpenTelemetry** and secure **OAuth** refresh

System Architecture and Platform Design - Spring Fest

Jul'24-Apr'25

- Scaled backend for **100M+ API requests** by deploying with **Nginx**, **Redis caching** on **4 AWS EC2 t3.large** instances
- Handled **80L+ transactions** by integrating **PhonePe API**, authentication using **JWT**, **OAuth2** and **rate-limited endpoints**
- Delivered **99.8% OTP/email delivery** using **MSG91**, **Mailer**, **async retry queues** and monitoring through **CloudWatch**
- Shipped **React Native app** on **Play Store** (**1K+ downloads**) with backend integration for **event access** and **admin tools**

Projects

Scalable Analytics Pipeline

Mar'25 - Apr'25

- Engineered **distributed** data processing pipeline using **Apache Spark** for multi-terabyte datasets with ETL and features
- Trained and optimized **ML models** (forecasting, clustering) with large-scale hyperparameter tuning and cross-validation
- Designed a **real-time streaming module** (Kafka + Spark Streaming) enabling low-latency analytics and predictive monitoring
- Deployed models as **REST APIs** and built an interactive dashboard with visualization and ad-hoc querying support

Conversational AI Platform for Employee Welfare

Feb'25 - Mar'25

- Built **AI-driven system** to monitor **employee well-being** by **behavioral signals**, **automated insights**, **LLM pipelines**
- Processed data from **1K+ users** using a **multi-stage pipeline** with **FastAPI**, **CRON jobs**, and **performance metrics**
- Flagged **high-risk users** with **42% improved accuracy** using **EMA-weighted scores** and **ensemble models**
- Orchestrated **LLM-powered chatbot** and **summaries** via **LangChain pipelines**, enabling **context-aware insights**

Skills & Expertise

- Programming Languages:** C/C++, Rust, Go, Elixir, Python, TypeScript, JavaScript, Bash, Awk, Verilog, SQL
- Libraries and Frameworks:** React, Next.js, React Native, Node.js, Express, Pandas, NumPy, GraphQL, Prisma, Django, Flask
- Tools/Platforms:** Docker, Kubernetes, Redis, MongoDB, Postgres, AWS, Azure, Git, Databricks, Linux Kernel, Raspberry Pi

Relevant Courses

- Computer Science:** Programming and Data Structures, Algorithm-I, Software Engineering, Computer Organization and Architecture, Compilers, Database Management Systems, Operating Systems, Computer Networks
- Mathematics:** Advanced Calculus, Linear Algebra and Numerical and Complex Analysis, Probability & Statistics, Discrete Structures, Formal Language and Automata Theory
- More Courses:** Basic Electronics, Switching Circuits and Logic Design, Machine Learning, Data Analytics

Awards and Achievements

- Codeforces** - sumitkumar-17 **1700** (Expert) and **LeetCode** - SumitKumar-17 **2181** (Guardian)
- Secured **AIR 2038** in JEE Advanced and **1870** in JEE Mains (1.2M candidates) in 2022.
- Contributed to open-source orgs **NixOS**, **Stripe**, and **Supabase** by writing **patches**, refining **Nix expressions**, and improving **system builds**
- Served as the Unit Leader for NSS Unit-4, coordinated activities, and received the **Best Volunteer Award**